1 Theoretical Underpinnings

1.1 Learning objectives

After completing this chapter, you will be able to:

- State Lotka's, Bradford's, and Zipf's laws and explain the empirical patterns they describe
- Test whether a corpus of publications approximately follows Lotka's inverse-square law
- Construct a Bradford plot to identify core and peripheral journals in a field
- Explain Price's law of exponential growth and the Matthew Effect in science
- Recognise power-law-like distributions in bibliometric data

1.2 Setup

```
library(tidyverse)
library(openalexR)
library(tidytext)
library(glue)

set.seed(20260509)

source(here::here("R", "api_helpers.R"))
source(here::here("R", "sci_palette.R"))
```

1.3 Conceptual background

Scientometrics rests on a small number of empirical regularities discovered across decades of studying scholarly communication:

Lotka's Law [@lotka1926] observes that the number of authors producing n papers is approximately proportional to $1/n^2$. A few prolific authors generate a large share of output, while most researchers publish only one or two papers.

Bradford's Law [@bradford1934] describes the scattering of articles across journals. For a given topic, a small core of journals contains roughly one-third of all relevant articles; a larger zone contains the next third; and a much larger periphery the final third. Each successive zone contains roughly the same number of articles but geometrically more journals.

Zipf's Law states that the frequency of a word is inversely proportional to its rank. In scientific text, a handful of terms dominate while the vast majority appear rarely.

Price's Law [@price1963] generalises these observations: the cumulative output of science grows exponentially, doubling roughly every 10–15 years. Price also noted that the square root of all contributors produce half of all contributions.

The **Matthew Effect** [@merton1968] — named after the Gospel of Matthew ("to those who have, more shall be given") — describes cumulative advantage in science. Well-known researchers attract disproportionate credit and resources, amplifying initial differences in visibility. This mechanism helps explain the skewed distributions observed by Lotka and Price.

1.4 Worked example

We use a corpus of works from the journal *Scientometrics* to demonstrate each law.

1.4.1 Fetching the data

```
works <- oa_fetch(
  entity = "works",
  primary_location.source.id = "S148561398",
  from_publication_date = "2015-01-01",
  to_publication_date = "2023-12-31",
  options = list(sample = 500, seed = 42)
)
```

1.4.2 Lotka's Law

We unnest authors and count the number of papers per author in this sample.

```
author_prod <- works |>
  select(id, authorships) |>
  unnest(authorships, names_sep = "_") |>
  count(authorships_display_name, name = "n_papers") |>
  count(n_papers, name = "n_authors") |>
  arrange(n_papers)

author_prod
```

```
#> # A tibble: 9 x 2
#>   n_papers n_authors
#>      <int>     <int>
#> 1        1      1144
#> 2        2        92
#> 3        3        24
#> 4        4         6
#> 5        5         2
#> 6        6         1
#> 7        7         2
#> 8        9         1
#> 9       12         1
```

```
lotka_c <- author_prod$n_authors[1]

author_prod |>
  ggplot(aes(x = n_papers, y = n_authors)) +
  geom_point(colour = palette_sci(1), size = 2) +
  geom_line(
    aes(y = lotka_c / n_papers^2),
    linetype = "dashed", colour = "grey40"
  ) +
  scale_x_log10() +
  scale_y_log10() +
  labs(x = "Papers per author", y = "Number of authors") +
  theme_sci()
```

The data approximate the inverse-square pattern, though real distributions often deviate from the exact exponent of 2.

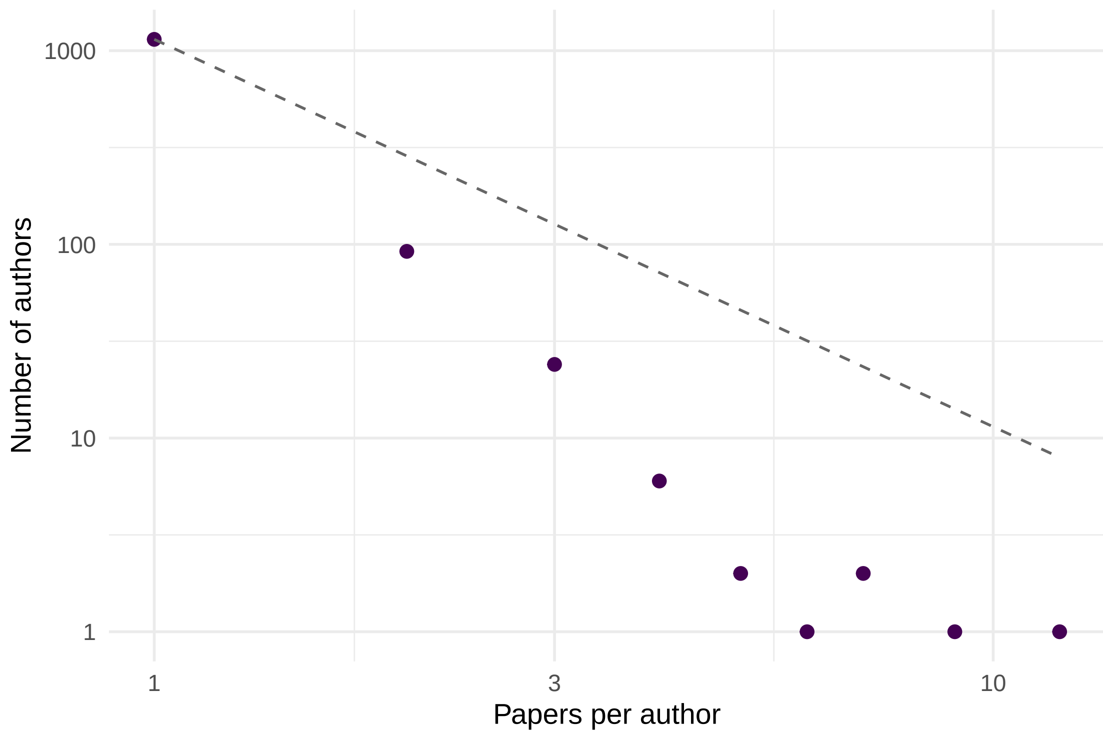

Figure 1: Author productivity distribution. The dashed line shows the inverse-square law predicted by Lotka.

1.4.3 Bradford's Law

We count how many articles each journal source contributes (across a broader query).

```r
broad_works <- oa_fetch(
  entity = "works",
  search = "bibliometrics",
  from_publication_date = "2020-01-01",
  to_publication_date = "2023-12-31",
  options = list(sample = 1000, seed = 42)
)

journal_counts <- broad_works |>
  filter(!is.na(source_display_name)) |>
  count(source_display_name, name = "articles", sort = TRUE) |>
  mutate(
    cum_articles = cumsum(articles),
    rank = row_number()
  )
```

```r
journal_counts |>
  ggplot(aes(x = rank, y = cum_articles)) +
  geom_line(colour = palette_sci(1), linewidth = 1) +
  scale_x_log10() +
  labs(x = "Journal rank (log scale)", y = "Cumulative articles") +
  theme_sci()
```

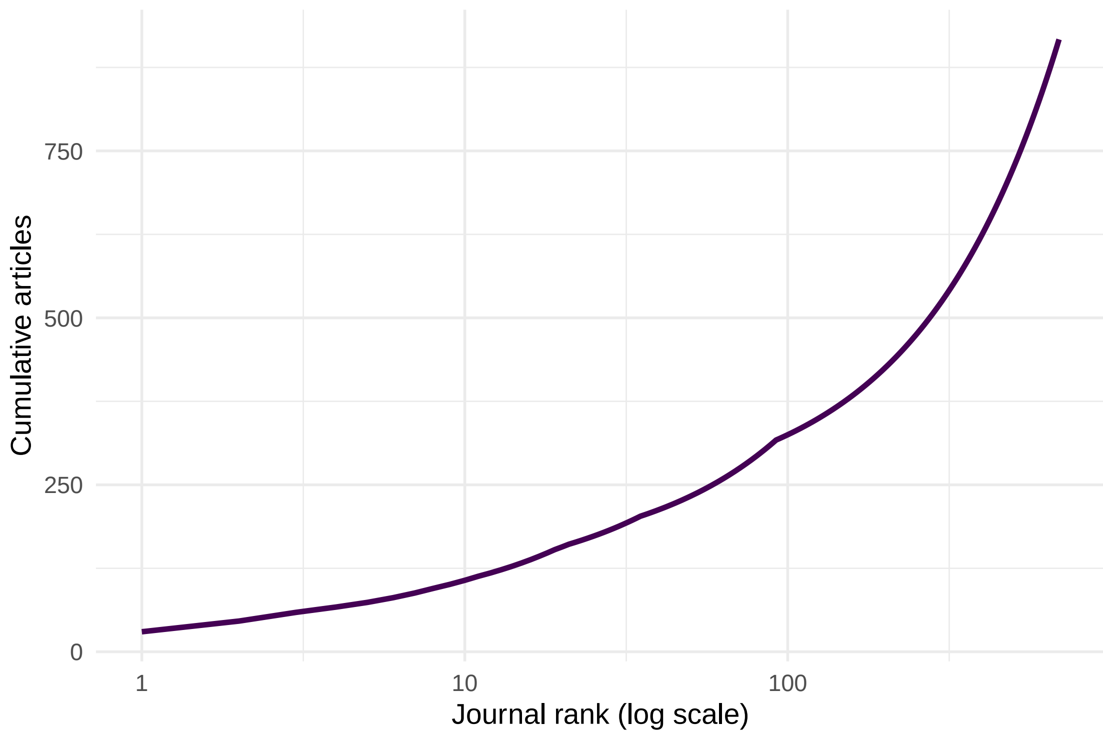

Figure 2: Bradford plot: cumulative articles vs. log journal rank.

1.4.4 Zipf's Law

We extract title words and examine their frequency distribution.

```
word_freq <- works |>
  select(display_name) |>
  unnest_tokens(word, display_name, token = "words") |>
  anti_join(tidytext::get_stopwords(), by = "word") |>
  count(word, sort = TRUE) |>
  mutate(rank = row_number())
```

```
word_freq |>
  head(200) |>
  ggplot(aes(x = rank, y = n)) +
  geom_point(colour = palette_sci(1), size = 1, alpha = 0.7) +
  scale_x_log10() +
  scale_y_log10() +
  labs(x = "Rank", y = "Frequency") +
  theme_sci()
```

1.4.5 Price's growth law

```
works |>
  mutate(year = year(publication_date)) |>
```

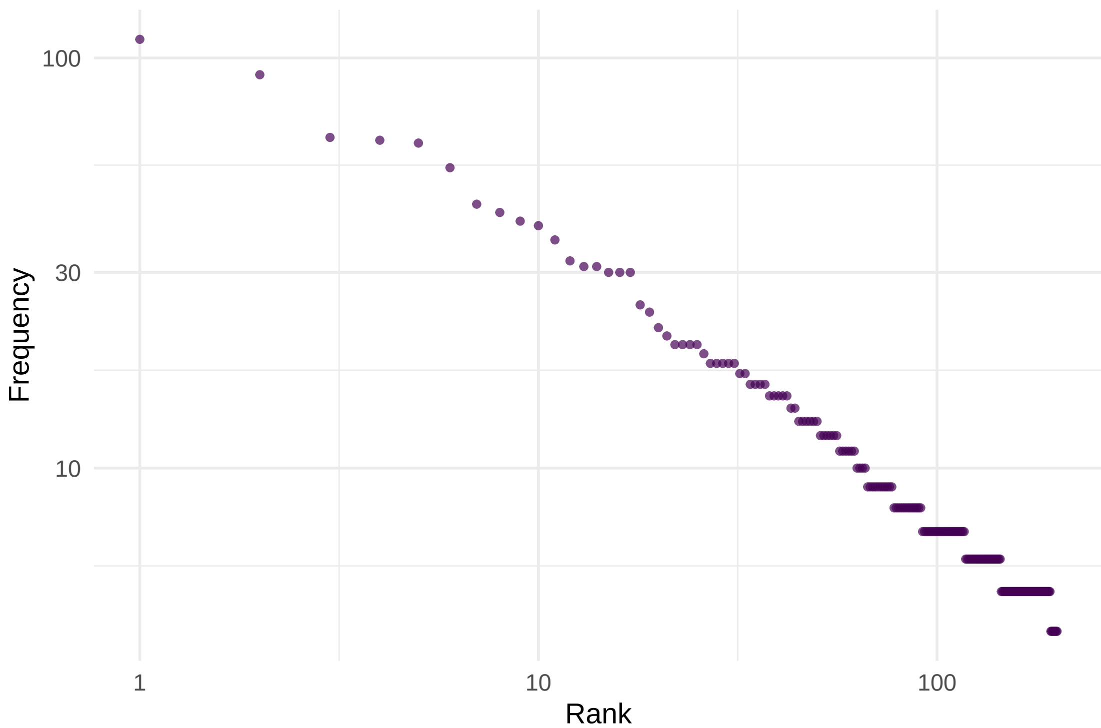

Figure 3: Word frequency vs. rank for article titles, approximating Zipf's law.

```
count(year) |>
ggplot(aes(x = year, y = n)) +
geom_col(fill = palette_sci(1)) +
labs(x = "Year", y = "Works in sample") +
theme_sci()
```

1.5 Diagnostics and interpretation

These laws are *empirical regularities*, not physical laws. They describe approximate patterns:

- Lotka's exponent is often not exactly 2; values between 1.5 and 3.5 are common depending on the field and time window.
- Bradford zones are sensitive to how you define the subject scope.
- Zipf's law holds best for common words; the tail deviates.
- Price's exponential growth eventually saturates — science cannot grow faster than the economy indefinitely.

The value of these laws is not predictive precision but the insight that **skewed distributions are the norm, not the exception**, in scholarly communication.

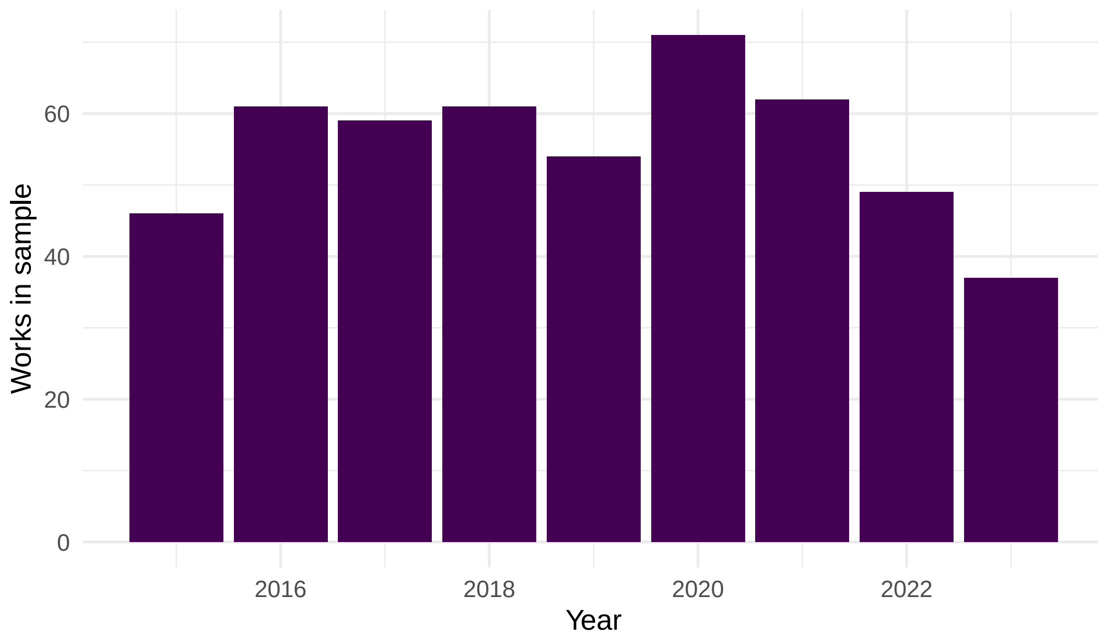

Figure 4: Annual publication counts for sampled works, illustrating the growth of bibliometrics literature.

1.6 Limitations and responsible use

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- These laws describe aggregate patterns, not individual behaviour. A researcher publishing one paper is not "less productive" in any evaluative sense — Lotka's law is a statistical observation, not a performance standard.
- The Matthew Effect [@merton1968] is a mechanism that *amplifies inequality*. Recognising it should prompt caution about using cumulative metrics (like the h-index) that inherit and reinforce this bias.
- Power-law fitting is notoriously difficult. Many distributions that "look" power-law on a log-log plot are better described by lognormal or stretched exponential models. Rigorous fitting requires maximum-likelihood methods and goodness-of-fit tests [@hicks2015].

1.8 Common pitfalls

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- **Eyeballing log-log plots.** A straight line on a log-log plot does not prove a power law. Use formal statistical tests.
- **Confusing description with prescription.** "Most authors publish few papers" does not mean they *should* publish more.
- **Ignoring field differences.** The exponent of Lotka's law and the scattering pattern of Bradford's law vary substantially across disciplines.
- **Assuming stationarity.** These patterns can shift over time as publication norms change (e.g., the rise of mega-journals).

1.10 Exercises

1. **Lotka exponent.** Fit a linear model to the log-log author productivity data. What exponent do you estimate? How does it compare to Lotka's original value of 2?

2. **Bradford zones.** Divide the journal list into three zones of roughly equal article counts. How many journals are in each zone? What is the ratio between successive zone sizes?

3. **Your field's Zipf distribution.** Fetch 500 works in a field of your choice. Extract title words and plot the Zipf distribution. Are the most frequent terms what you would expect?

4. **Matthew Effect simulation.** Write a simple simulation: start 100 authors with 1 paper each. At each time step, an author gains a new paper with probability proportional to their current count. After 50 steps, plot the distribution. Does it resemble Lotka's law?

1.11 Solutions

Solutions are provided in 1.11.

1.12 Further reading

- @lotka1926 — The original observation on author productivity distributions.
- @bradford1934 — Journal scattering and the Bradford law.
- @price1963 — Exponential growth of science and cumulative advantage.
- @merton1968 — The Matthew Effect: how advantage accumulates in science.
- @garfield1955 — Citation indexing, the empirical basis for much of scientometrics.
- @waltman2016 — A modern review contextualising these classical laws within citation impact measurement.

1.13 Session info

```r
sessionInfo()
```

```
#> R version 4.4.1 (2024-06-14)
#> Platform: x86_64-pc-linux-gnu
#> Running under: Ubuntu 24.04.4 LTS
#>
#> Matrix products: default
#> BLAS:   /usr/lib/x86_64-linux-gnu/openblas-pthread/libblas.so.3
#> LAPACK: /usr/lib/x86_64-linux-gnu/openblas-pthread/libopenblasp-r0.3.26.so;  LAPACK version 3.12.0
#>
#> locale:
#>  [1] LC_CTYPE=C.UTF-8       LC_NUMERIC=C           LC_TIME=C.UTF-8
#>  [4] LC_COLLATE=C.UTF-8     LC_MONETARY=C.UTF-8    LC_MESSAGES=C.UTF-8
#>  [7] LC_PAPER=C.UTF-8       LC_NAME=C              LC_ADDRESS=C
#> [10] LC_TELEPHONE=C         LC_MEASUREMENT=C.UTF-8 LC_IDENTIFICATION=C
#>
#> time zone: UTC
#> tzcode source: system (glibc)
#>
#> attached base packages:
```

```
#> [1] stats     graphics  grDevices utils     datasets  methods   base
#>
#> other attached packages:
#>  [1] tidytext_0.4.3  glue_1.8.1      openalexR_3.0.1 lubridate_1.9.5
#>  [5] forcats_1.0.1   stringr_1.6.0   dplyr_1.2.1     purrr_1.2.2
#>  [9] readr_2.2.0     tidyr_1.3.2     tibble_3.3.1    ggplot2_4.0.3
#> [13] tidyverse_2.0.0 bookdown_0.46   fs_2.1.0        rmarkdown_2.31
#>
#> loaded via a namespace (and not attached):
#>  [1] janeaustenr_1.0.0  viridis_0.6.5      utf8_1.2.6         generics_0.1.4
#>  [5] lattice_0.22-6     stringi_1.8.7      hms_1.1.4          digest_0.6.39
#>  [9] magrittr_2.0.5     evaluate_1.0.5     grid_4.4.1         timechange_0.4.0
#> [13] RColorBrewer_1.1-3 fastmap_1.2.0      Matrix_1.7-0       rprojroot_2.1.1
#> [17] jsonlite_2.0.0     tinytex_0.59       gridExtra_2.3      stopwords_2.3
#> [21] httr_1.4.8         viridisLite_0.4.3  scales_1.4.0       codetools_0.2-20
#> [25] cli_3.6.6          rlang_1.2.0        tokenizers_0.3.0   withr_3.0.2
#> [29] yaml_2.3.12        otel_0.2.0         tools_4.4.1        tzdb_0.5.0
#> [33] here_1.0.2         curl_7.1.0         vctrs_0.7.3        R6_2.6.1
#> [37] lifecycle_1.0.5    pkgconfig_2.0.3    pillar_1.11.1      gtable_0.3.6
#> [41] Rcpp_1.1.1-1.1     xfun_0.57          tidyselect_1.2.1   knitr_1.51
#> [45] dichromat_2.0-0.1  farver_2.1.2       SnowballC_0.7.1    htmltools_0.5.9
#> [49] labeling_0.4.3     compiler_4.4.1     S7_0.2.2
```